

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

(1) Let  $m$  be the Lebesgue measure on  $[0, 1]$  and  $I_A(x)$  denote the indicator function of  $A$ .

(a) Consider for  $n \geq 1$ ,  $f_n(x) = nI_{[0, \frac{1}{n}]}(x)$  and  $f(x) = 0$  for  $x \in [0, 1]$ . Show:

(i)  $f_n \rightarrow f$  a.e.( $m$ ).

(ii)  $f_n$  is **not** uniformly integrable (UI).

(iii) Check  $f_n \rightarrow f$  in  $L^1$ .

(b) In general, under what conditions on the functions  $\{f_n\}$  and  $f$ , would

" $f_n \rightarrow f$  a.e. ( $m$ )" imply " $f_n \rightarrow f$  in  $L^p$  ( $0 < p < \infty$ )"?

(c) Show that  $f_n \rightarrow f$  in  $L^p$  implies that  $f_n \rightarrow f$  in  $L^q$ , for any  $0 < q < p < \infty$ .

((3×6)+6+8=32 points)

(2) Suppose  $X$  is a random variable defined on some probability space  $(\Omega, \mathcal{F}, P)$ .

(a) If  $M(t) = E(e^{tX})$ ,  $t \in \mathbb{R}$  is the moment generating function of  $X$ , then show

$$P(X > a) \leq \inf_{t>0} e^{-ta} M(t), \quad a > 0.$$

(b) If  $X \geq 0$  is random variable with  $\text{Var}(X) < \infty$ . Prove:

$$P(X > 0) \geq \frac{[E(X)]^2}{E(X^2)}.$$

[Hint: Use Cauchy-Schwarz Inequality]

(8+10=18 points)

(3) (a) State precisely the Dominated convergence Theorem (DCT) (Not the EDCT!).

(b) Prove that the conclusion of DCT remains true even if " $f_n \rightarrow f$  a.e. ( $\mu$ )" is replaced by " $f_n \rightarrow f$  in measure" in the statement.

(6 + 9 = 15 points)

(4) Let  $(\Omega, \mathcal{F})$  be a measurable space, and  $\mu, \mu_1, \mu_2, \nu$  are  $\sigma$ -finite measures on it. Define the set function:  $\mu_3 \equiv \mu_1 + \mu_2$  (i.e., for all  $A \in \mathcal{F}$ ,  $\mu_3(A) = \mu_1(A) + \mu_2(A)$ ). Then, prove the following:

(a)  $\mu_3$  is a measure.

(b) If  $\mu_1 \ll \nu$  and  $\mu_2 \ll \nu$ , then  $\mu_3 \ll \nu$ .

(c) Express the Radon-Nikodym derivative  $\frac{d\mu_3}{d\nu}$  in terms of  $\frac{d\mu_1}{d\nu}$  and  $\frac{d\mu_2}{d\nu}$  (proof required).

(d) If  $\mu \ll \nu$  and  $\mu \perp \nu$  holds, show that  $\mu$  is the zero measure.

(e) If  $\mu_1 = \text{Lognormal}(5, 6)$ ,  $\mu_2 = \text{Geometric}(0.3)$  and  $\nu = \text{Poisson}(8)$  on  $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ , then find the Lebesgue decomposition of  $\mu_3$  w.r.t.  $\nu$ , and find the Radon Nikodym derivative  $\frac{d\mu_{3a}}{d\nu}$ , where  $\mu_{3a}$  is the absolutely continuous component of  $\mu_3$  in the decomposition.

(6+8+8+8+10=40 points)

TABLE 4.6.1. Some discrete univariate distributions.

| Mean $\mu$                                            | $\mu(A), A \in \mathcal{B}(\mathbb{R})$                              |
|-------------------------------------------------------|----------------------------------------------------------------------|
| Bernoulli ( $p$ ),<br>$0 < p < 1$                     | $\sum_{i=0}^1 p^i (1-p)^{1-i} I_A(i)$                                |
| Binomial ( $n, p$ ),<br>$0 < p < 1, n \in \mathbb{N}$ | $\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$                   |
| Geometric ( $p$ ),<br>$0 < p < 1$                     | $\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$                                |
| Poisson ( $\lambda$ ),<br>$0 < \lambda < \infty$      | $\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$ |

TABLE 4.6.2. Some standard absolutely continuous distributions. Here  $m$  denotes the Lebesgue measure on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ .

| Measure $\mu$                                                                        | $\mu(A), A \in \mathcal{B}(\mathbb{R})$                                                                                                                                       |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Uniform ( $a, b$ ),<br>$-\infty < a < b < \infty$                                    | $m(A \cap [a, b]) / (b - a)$                                                                                                                                                  |
| Exponential ( $\beta$ ),<br>$\beta \in (0, \infty)$                                  | $\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$                                                                                                              |
| Gamma ( $\alpha, \beta$ ),<br>$\alpha, \beta \in (0, \infty)$                        | $\int_A \frac{1}{\Gamma(\alpha) \beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$<br>where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx, a \in (0, \infty)$ |
| Beta ( $\alpha, \beta$ ),<br>$\alpha, \beta \in (0, \infty)$                         | $\frac{1}{B(\alpha, \beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0,1)}(x) m(dx),$<br>where $B(a, b) = \Gamma(a) \Gamma(b) / \Gamma(a+b), a, b \in (0, \infty)$             |
| Cauchy ( $\gamma, \sigma$ )<br>$\gamma \in \mathbb{R}, \sigma \in (0, \infty)$       | $\int_A \frac{1}{\pi \sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$                                                                                                  |
| Normal ( $\gamma, \sigma^2$ ),<br>$\gamma \in \mathbb{R}, \sigma \in (0, \infty)$    | $\int_A \frac{1}{\sqrt{2\pi\sigma}} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$                                                                                                    |
| Lognormal ( $\gamma, \sigma^2$ ),<br>$\gamma \in \mathbb{R}, \sigma \in (0, \infty)$ | $\int_A \frac{1}{\sqrt{2\pi\sigma}} \frac{e^{-(\log x - \gamma)^2 / 2\sigma^2}}{x} I_{(0, \infty)}(x) m(dx)$                                                                  |

1. (a) (i) For  $x = 1$ ,  $f_n(1) = f(1) = 0$ . So  $f_n(1) \rightarrow f(1)$  as  $n \rightarrow \infty$ .  
 For  $x = 0$ ,  $f(0) = 0$  and  $f_n(0) = n$ ,  $\forall n$ . So  $f_n(0) \nrightarrow f(0)$  as  $n \rightarrow \infty$ .  
 For  $x \in (0, 1)$ , then  $\exists n_0$  s.t.  $\frac{1}{n_0} < x$  and  $\forall n \geq n_0$ ,  $f_{n_0}(x) = 0 = f(x)$ . So  $f_n(x) \rightarrow f(x)$ ,  $x \in (0, 1)$  as  $n \rightarrow \infty$ .  
 So  $\{x : f_n(x) \nrightarrow f(x)\} = \{0\}$  and  $m(\{0\}) = 0$ . Hence  $f_n \rightarrow f$  a.e. ( $\mu$ ).
- (ii) Since  $\int |f_n| I_{\{|f_n| > T\}} d\mu = nm([0, \frac{1}{n})) = 1$ ,  $\forall n > T$ , then  $\sup_{n \geq 1} \int |f_n| I_{\{|f_n| > T\}} d\mu \geq 1$ . So  $\lim_{T \rightarrow \infty} \sup_{n \geq 1} \int |f_n| I_{\{|f_n| > T\}} d\mu \neq 0$ . Hence  $f_n$  is not UI.
- (iii)  $\int |f_n - f| d\mu = \int n I_{[0, \frac{1}{n})} d\mu = n \cdot \frac{1}{n} = 1 \nrightarrow 0$ . So  $f_n \nrightarrow f$  in  $L^1$ .
- (b) If  $\{f_n^p\}$  is UI, i.e.  $\lim_{T \rightarrow \infty} \sup_{n \geq 1} \int |f_n| I_{\{|f_n| > T\}} d\mu = 0$ , then  $f_n \rightarrow f$  a.e. ( $m$ ) would imply  $f_n \rightarrow f$  in  $L^p$  (HW problem: 2.36(b)) since  $m(\Omega) = m([0, 1]) = 1 < \infty$ . There are other correct answers too, for example, along the lines of DCT.
- (c) Note that in this case  $r = p/q > 1$  and hence  $\phi(x) = x^r$ ,  $x \in [0, 1]$  is a convex function. Using Jensen's, we have

$$\left( \int |f_n - f|^q d\mu \right)^{\frac{p}{q}} \leq \left( \int |f_n - f|^{q \frac{p}{q}} d\mu \right) = \left( \int |f_n - f|^p d\mu \right).$$

or in other words,  $\|f_n - f\|_q \leq \|f_n - f\|_p \rightarrow 0$ , as  $n \rightarrow \infty$ .

2. (a) Using Markov's inequality, for each  $t > 0$ ,

$$P(X > a) = P(e^{tX} > e^{ta}) \leq e^{-ta} M(t),$$

and hence the result follows.

- (b) Apply Cauchy-Schwarz to  $X I_{\{X > 0\}}$  to obtain

$$E[X I_{\{X > 0\}}] \leq [E(X^2)]^{1/2} [P(X > 0)]^{1/2}$$

using the fact that  $I_{\{X > 0\}}^2 = I_{\{X > 0\}}$  and  $E[I_{\{X > 0\}}] = P(X > 0)$ . Also, since  $X$  is nonnegative,  $E(X) = E(X I_{\{X=0\}}) + E(X I_{\{X>0\}}) = E(X I_{\{X>0\}})$ . Hence, squaring both sides above, we get  $[E(X)]^2 \leq E(X^2) P(X > 0)$  and the result follows.

3. (a) Let  $(\Omega, \mathcal{F}, \mu)$  be a measure space and  $f_n, f, g$  be measurable functions. If  $|f_n| \leq g$ , a.e. ( $\mu$ ),  $\forall n \geq 1$ ,  $\int g d\mu < \infty$  and  $f_n \rightarrow f$  a.e. ( $\mu$ ), then  $f \in L^1(\Omega, \mathcal{F}, \mu)$ ,  $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$  and  $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$ .
- (b) We need to prove the following modified DCT:  
 Let  $(\Omega, \mathcal{F}, \mu)$  be a measure space and  $f_n, f, g$  be measurable functions. If  $|f_n| \leq g$ , a.e. ( $\mu$ ),  $\forall n \geq 1$ ,  $\int g d\mu < \infty$  and  $f_n \rightarrow f$  in measure, then  $f \in L^1(\Omega, \mathcal{F}, \mu)$ ,  $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$  and  $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$ .

Proof: First we state two facts in real analysis:

Fact 1: A sequence  $\{x_n\}$  converges to  $x$  iff for any subsequence  $\{x_{n_k}\}$  of  $\{x_n\}$ , there is a further subsequence  $\{x_{n_{k_l}}\}$  of  $\{x_{n_k}\}$  such that  $x_{n_{k_l}} \rightarrow x$ .

Fact 2 (Theorem 2.5.2):  $f_n \rightarrow f$  in measure  $\Rightarrow$  there exists a subsequence  $\{f_{n_k}\}$  such that  $f_{n_k} \rightarrow f$  a.e. ( $\mu$ ).

Then we prove  $f \in L^1(\Omega, \mathcal{F}, \mu)$  and  $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$ .

Denote  $a_n = \int f_n d\mu$ ,  $a = \int f d\mu$  and  $b_n = \int |f_n - f| d\mu$ . To show  $a_n \rightarrow a$  and  $b_n \rightarrow 0$ , we need to use Fact 1. Choose any subsequence  $\{a_{n_k}\}$  of  $\{a_n\}$ , and for this subsequence, choose the corresponding subsequence functions  $\{f_{n_k}\}$  of  $\{f_n\}$ . Since  $f_n \rightarrow f$  in measure, then  $\lim_{n \rightarrow \infty} \mu(|f_n - f| > \varepsilon) = 0$ ,  $\forall \varepsilon > 0$ , thus  $\lim_{k \rightarrow \infty} \mu(|f_{n_k} - f| > \varepsilon) = 0$ ,  $\forall \varepsilon > 0$ , i.e.  $f_{n_k} \rightarrow f$  in measure. By Fact 2, we can get a sub-subsequence  $f_{n_{k_l}} \rightarrow f$  a.e. ( $\mu$ ). For this sub-subsequence  $\{f_{n_{k_l}}\}$ , conditions of DCT are true. Hence from DCT, we have  $f \in L^1(\Omega, \mathcal{F}, \mu)$  and  $\lim_{l \rightarrow \infty} \int f_{n_{k_l}} d\mu = \int f d\mu$ , i.e.  $a_{n_{k_l}} \rightarrow a$ . By Fact 1, we get  $a_n \rightarrow a$ , i.e.  $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$ . Similarly, we have  $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$ .

4. (a)  $\mu_3(\emptyset) = \mu_1(\emptyset) + \mu_2(\emptyset) = 0$ ;  
 $\mu_3(A) = \mu_1(A) + \mu_2(A) \geq 0 \Rightarrow \mu_3(A) \in [0, \infty]$ ,  $\forall A \in \mathcal{F}$ ;  
 If  $\{A_i\}$  is a disjoint collection of sets in  $\mathcal{F}$ , then  $\mu_3(\cup_i A_i) = \mu_1(\cup_i A_i) + \mu_2(\cup_i A_i) = \sum_i \mu_1(A_i) + \sum_i \mu_2(A_i) = \sum_i (\mu_1(A_i) + \mu_2(A_i)) = \sum_i \mu_3(A_i)$ .  
 So  $\mu_3$  is a measure.
- (b)  $\nu(A) = 0 \Rightarrow \mu_1(A) = \mu_2(A) = 0 \Rightarrow \mu_3(A) = \mu_1(A) + \mu_2(A) = 0$ . So  $\mu_3 \ll \nu$ .
- (c) Claim:  $\frac{d\mu_3}{d\nu} = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$  a.e. ( $\nu$ ). *Proof:* Since  $\mu_1 \ll \nu$  and  $\mu_2 \ll \nu$ , then  $\exists f_1, f_2 \geq 0$  s.t.  $\mu_1(A) = \int_A f_1 d\nu$ ,  $\mu_2(A) = \int_A f_2 d\nu$ . By definition,  $\mu_3(A) = \mu_1(A) + \mu_2(A) = \int_A f_1 d\nu + \int_A f_2 d\nu = \int_A (f_1 + f_2) d\nu$  by linearity of integrals. Hence by uniqueness of RN derivatives,  $\frac{d\mu_3}{d\nu} = f_1 + f_2 = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$  a.e. ( $\nu$ ).
- (d)  $\mu \perp \nu \Rightarrow \exists B \in \mathcal{F}$  s.t.  $\nu(B) = \mu(B^c) = 0$ . Since  $\mu \ll \nu$  and  $\nu(B) = 0$ , then  $\mu(B) = 0$ . So  $\mu(B) = \mu(B^c) = 0 \Rightarrow \mu(\Omega) = 0$ . Thus  $\forall A \in \mathcal{F}$ ,  $0 \leq \mu(A) \leq \mu(\Omega) = 0 \Rightarrow \mu(A) \equiv 0$ ,  $\forall A \in \mathcal{F}$ , i.e.  $\mu \equiv 0$ .
- (e) Let  $B = \mathbb{N} \cup \{0\} = \mathbb{N}_0$ , then  $\mu_1(B) = 0 = \nu(B^c)$ , i.e.  $\mu_1 \perp \nu$ . Choose  $\mu_s = \mu_1$ ,  $\mu_a = \mu_2$  and  $\mu_3 = \mu_a + \mu_s$ . To show  $\mu_a \ll \nu$  and compute  $\frac{d\mu_a}{d\nu}$ .  
 Let  $\mu_c(A) = |A \cap \mathbb{N}_0|$ ,  $\forall A$  which defines a counting measure on  $\mathbb{N}_0$ . From table 4.6.1,  $\nu(A) = 0 \Rightarrow A \cap \mathbb{N}_0 = \emptyset \Rightarrow \mu_a(A) = 0$ , i.e.  $\mu_a \ll \nu$ . So  $\mu_a$  is the absolute continuous part of  $\mu_3$  w.r.t.  $\nu$ . Then  $\mu_2(A) = \sum_{x \in \mathbb{N}_0} I_A(x)(0.7)^x 0.3 = \int p_G(x) d\mu_c(x)$  where  $p_G(x) = I_{\mathbb{N}_0}(x)(0.7)^x 0.3$ . Similarly  $\nu(A) = \int p_P(x) d\mu_c(x)$  where  $p_P(x) = e^{-8} 8^x (x!)^{-1} I_{\mathbb{N}_0}(x)$ .

Since  $p_P(x) > 0$  a.e.  $(\mu_c)$ , then

$$\begin{aligned}\frac{d\mu_a}{d\nu}(x) &= \frac{d\mu_2}{d\mu_c} \left( \frac{d\nu}{d\mu_c} \right)^{-1} \\ &= \left( I_{\mathbb{N}_0}(x)(0.7)^x 0.3 \right) \cdot \left( I_{\mathbb{N}_0}(x) e^{-8} 8^x (x!)^{-1} \right)^{-1} \\ &= \frac{(0.7)^x 0.3 e^8 x!}{8^x} I_{\mathbb{N}_0}(x), \text{ a.e. } (\nu).\end{aligned}$$



## Midterm 2 (2014)

① (a)

(i) Consider when

$$x=1 : \text{ then } f_n(1) = n \cdot \mathbb{I}_{[0, \frac{1}{n})}(1) = 0 \quad \forall n \geq 1$$

$$x=0 : \text{ then } f_n(0) = n \cdot \mathbb{I}_{[0, \frac{1}{n})}(0) = n \quad \forall n \geq 1$$

$$x \in (0, 1) : \text{ then } \forall x \exists \text{ some } n_0 \text{ s.t. } \frac{1}{n_0} < x \Rightarrow$$

$$f_{n_0}(x) = n_0 \cdot \mathbb{I}_{[0, \frac{1}{n_0})}(x) = 0, \text{ so}$$

$$\lim_{n \rightarrow \infty} f_n(x) = 0 \quad \forall x \in (0, 1).$$

Thus,

$$\lim_{n \rightarrow \infty} f_n(x) = 0 \quad \forall x \in (0, 1]$$

$$\text{and } m(\{0\}) = 0, \text{ so } f_n \rightarrow f \text{ a.e. (m).}$$

(ii) Consider for some  $T > 0$

$$\begin{aligned} a_n(T) &= \int |f_n| \mathbb{I}_{[f_n > T]} dm \\ &= \int n \mathbb{I}_{[0, \frac{1}{n})} \mathbb{I}_{[f_n > T]} dm \end{aligned}$$

When  $n > T$ , then

$$= \int n \mathbb{I}_{[0, \frac{1}{n})} dm$$

$$= n m([0, \frac{1}{n}))$$

$$= n \left( \frac{1}{n} - 0 \right)$$

$$= 1$$

Then

$$\sup_{n \geq 1} a_n(T)$$

$$= \sup_{n \geq 1} = 1$$

Hence,  $\sup_{n \geq 1} a_n(T) \not\rightarrow 0$

as  $t \rightarrow \infty$ . Hence, not U.I.

(iii) Consider that

$$\begin{aligned}\lim_{n \rightarrow \infty} \int |f_n - f| dm &= \lim_{n \rightarrow \infty} \int |n \mathbb{I}_{[0, \frac{1}{n}]} - 0| dm \\&= \lim_{n \rightarrow \infty} \int n \mathbb{I}_{[0, \frac{1}{n}]} dm \\&= \lim_{n \rightarrow \infty} (n m([0, \frac{1}{n}])) \\&= \lim_{n \rightarrow \infty} n \left( \frac{1}{n} - 0 \right) \\&= \lim_{n \rightarrow \infty} 1 \\&= 1 \\&\neq 0\end{aligned}$$

Hence,  $f_n \rightarrow f$  in  $L^p$ .

(b) When  $\{f_n\}^p$  is UI and  $P(\Omega) < \infty$ .

(c) Let  $f_n \rightarrow f$  in  $L^p$ . Then

$$\lim_{n \rightarrow \infty} \int |f_n - f|^p dP = 0.$$

WTS that this implies  $f_n \rightarrow f$  in  $L^q$  for  $0 < q < p < \infty$ .

Let  $\phi(x) = x^r$  where  $r > 1$ . Then  $\phi(x)$  is a convex function, and by Jensen's inequality

$$\phi\left(\int x dP\right) \leq \int \phi(x) dP.$$

Let  $x = |f_n - f|^q$ . Then

$$\left(\int |f_n - f|^q dP\right)^r \leq \int (|f_n - f|^q)^r dP$$



Consider that since  $q < p \Rightarrow \frac{p}{q} > 1$ . Hence

$$\left( \int |f_n - f|^q dP \right)^{\frac{p}{q}} \leq \int (|f_n - f|^q)^{\frac{p}{q}} dP \\ = \int |f_n - f|^p dP.$$

By hypothesis, we have that

$$\lim_{n \rightarrow \infty} \int |f_n - f|^p dP = 0.$$

Thus,

$$\lim_{n \rightarrow \infty} \int |f_n - f|^q dP = 0 \quad \text{as well.}$$

3)  $X$  a r.v. on  $(\Omega, \mathcal{F}, P)$ . Then

(a)  $M(t) = E[e^{tx}]$  w/  $t \in \mathbb{R}$ . WTS

$$P(X > a) \leq \inf_{t > 0} e^{-ta} M(t) \quad a > 0$$

By Cramer's inequality,

$$P(X > a) \leq \inf_{t > 0} \frac{E[e^{tx}]}{e^{ta}} = \inf_{t > 0} e^{-ta} M(t)$$

for  $a > 0$ . (Or could also apply Markov's Inequality)

(b)

(3) (a) DCT: Let  $f_n, f$ , and  $g$  be measurable functions on the measure space  $(\Omega, \mathcal{F}, \mu)$ . Suppose that  $|f_n| \leq g \quad \forall n \geq 1 \text{ a.e. } (\mu)$ ,  $\int g d\mu < \infty$ , and  $f_n \rightarrow f \text{ a.e. } (\mu)$ . Then,

$$(i) f \in L^1(\Omega, \mathcal{F}, \mu) \quad \left( \int |f| d\mu < \infty \right)$$

$$(ii) \lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$$

$$(iii) \lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0.$$

(b) Want to show that the conclusion of DCT remains true if  $f_n \xrightarrow{m} f$  instead of  $f_n \rightarrow f \text{ a.e. } (\mu)$ . That is, we want to show that if  $f, f_n$ , and  $g$  are measurable functions on the measure space  $(\Omega, \mathcal{F}, \mu)$  with

$$(1) |f_n| \leq g \text{ a.e. } (\mu) \quad \forall n \geq 1,$$

$$(2) \int g d\mu < \infty, \text{ and}$$

$$(3) f_n \xrightarrow{m} f,$$

then

$$(1) f \in L^1(\Omega, \mathcal{F}, \mu)$$

$$(2) \lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$$

$$(3) \lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0.$$

Proof

Let  $a_n = \int f_n d\mu$  be a sequence for  $n \geq 1$ , and let  $a = \int f d\mu$ . We want to show that  $a_n \rightarrow a$ .

For any subsequence  $\{a_{n_k}\}_{k \geq 1}$  of  $\{a_n\}_{n \geq 1}$ , choose the corresponding subsequence of functions  $\{f_{n_k}\}_{k \geq 1}$  of  $\{f_n\}_{n \geq 1}$ . We know that  $f_n \xrightarrow{m} f$ . Let us show that  $f_{n_k} \xrightarrow{m} f$ . We have that

$$f_n \xrightarrow{m} f \Rightarrow \lim_{n \rightarrow \infty} \mu(|f_n - f| > \varepsilon) = 0 \quad \forall \varepsilon > 0.$$

Therefore,

$$\lim_{k \rightarrow \infty} \mu(|f_{n_k} - f| > \varepsilon) = 0 \quad \forall \varepsilon > 0 \text{ as well.}$$

That is,  $f_{n_k} \xrightarrow{m} f$ . Since  $f_{n_k} \xrightarrow{m} f$ ,  $\exists$  some subsequence  $\{f_{n_{k_l}}\}_{l \geq 1}$  s.t.  $f_{n_{k_l}} \rightarrow f$  a.e. ( $\mu$ ). Now, the conditions of the DCT hold for  $\{f_{n_{k_l}}\}_{l \geq 1}$ .

That is,

$$(1) f \in L^1(\Omega, \mathcal{F}, \mu),$$

$$(2) \lim_{l \rightarrow \infty} \int f_{n_{k_l}} d\mu = \int f d\mu, \text{ and}$$

$$(3) \lim_{l \rightarrow \infty} \int |f_{n_{k_l}} - f| d\mu = 0.$$

This proves the first part of the modified DCT ( $f \in L^1(\Omega, \mathcal{F}, \mu)$ ). ✓ A real analysis fact says that a sequence  $a_n \rightarrow a$  iff for all subsequences  $a_{n_k}$ ,  $\exists$  some sub-subsequence  $a_{n_{k_l}}$  s.t.  $a_{n_{k_l}} \rightarrow a$ . In our case, let  $a_{n_{k_l}} = \int f_{n_{k_l}} d\mu$ . We have shown that for any  $\{a_{n_k}\}_{k \geq 1}$ ,  $\exists$  some

subsequence  $\{a_{n_k}\}_{k=1}^\infty$  such that

$$a_{n_k} \rightarrow a.$$

Thus, by the real analysis fact,  $a_n \rightarrow a$ . That is,

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu.$$

Now let  $b_n = \int |f_n - f| d\mu$ . We want to show that  $b_n \rightarrow 0$ . For any subsequence,  $\{b_{n_k}\}_{k=1}^\infty$  where

$$b_{n_k} = \int |f_{n_k} - f| d\mu,$$

$\exists$  a sub-subsequence  $\{b_{n_{k_l}}\}_{l=1}^\infty$  s.t.

$$b_{n_{k_l}} = \int |f_{n_{k_l}} - f| d\mu$$

with  $b_{n_{k_l}} \rightarrow 0$  (we just showed this). Then we can apply the real analysis fact again to say that  $b_n \rightarrow 0$ . That is

$$\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0. \quad \checkmark$$

④  $(\Omega, \mathcal{F})$  a measurable space and  $\mu_1, \mu_2, \mu_3$  are  $\sigma$ -finite measures where  $\mu_3 \equiv \mu_1 + \mu_2$ .

⊙ Want to prove that  $\mu_3$  is a measure.

$$(i) \mu_3(\emptyset) = \mu_1(\emptyset) + \mu_2(\emptyset) = 0 + 0 = 0 \quad \checkmark$$

$$(ii) \mu_3(A) \in [0, \infty) \quad \forall A \in \mathcal{F}. \quad \text{Since } \mu_1 \text{ and } \mu_2$$

are measures

$$\mu_1(A) \in [0, \infty) \quad \text{for all } A \in \mathcal{F}$$

$$\mu_2(A) \in [0, \infty) \quad \text{for all } A \in \mathcal{F}.$$

Thus,

$$\mu_1(A) + \mu_2(A) \in [0, \infty). \quad \checkmark$$

(iii) Let  $A_1, A_2, \dots \in \mathcal{F}$  be a collection of disjoint sets with  $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$ . Want to show that

$$\mu_3\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_3(A_n).$$

Consider that

$$\begin{aligned} \mu_3\left(\bigcup_{n=1}^{\infty} A_n\right) &= \mu_1\left(\bigcup_{n=1}^{\infty} A_n\right) + \mu_2\left(\bigcup_{n=1}^{\infty} A_n\right) \\ &= \sum_{n=1}^{\infty} \mu_1(A_n) + \sum_{n=1}^{\infty} \mu_2(A_n) \\ &= \sum_{n=1}^{\infty} \mu_1(A_n) + \mu_2(A_n) \\ &= \sum_{n=1}^{\infty} \mu_3(A_n). \end{aligned} \quad \checkmark$$

Hence,  $\mu_3$  is a measure.

(b) If  $\mu_1 \ll \nu$  and  $\mu_2 \ll \nu$ , then for  $A \in \mathcal{F}$

$$\nu(A) = 0 \Rightarrow \mu_1(A) = 0 = \mu_2(A)$$

$$\Rightarrow \mu_3(A) = \mu_1(A) + \mu_2(A) = 0 + 0 = 0.$$

Hence,  $\nu(A) = 0 \Rightarrow \mu_3(A) = 0$

$$\Rightarrow \mu_3 \ll \nu.$$

$\checkmark$

(c) We want to express  $\frac{d\mu_3}{d\nu}$  in terms of  $\frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$ .

Consider that  $\mu_1 \ll \nu \Rightarrow \exists$  some non-negative measurable function  $h_1$  on  $(\Omega, \mathcal{F})$  such that

$$\mu_1(A) = \int_A h_1 d\nu \quad \forall A \in \mathcal{F} \quad \text{w/ } h_1 = \frac{d\mu_1}{d\nu}.$$

Similarly,  $\mu_2 \ll \nu \Rightarrow \exists h_2 \xleftarrow{\text{s.t.}} \text{non-neg. meas. on } (\Omega, \mathcal{F})$

$$\mu_2(A) = \int_A h_2 d\nu \quad \forall A \in \mathcal{F} \quad \text{w/ } h_2 = \frac{d\mu_2}{d\nu}.$$

Now,

$$\mu_3(A) = \mu_1(A) + \mu_2(A)$$

$$= \int_A h_1 d\nu + \int_A h_2 d\nu$$

$$= \int_A (h_1 + h_2) d\nu \quad (\text{by linearity of integrals})$$

$$= \int_A h_3 d\nu$$

where  $h_3 = h_1 + h_2 = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$  is a non-neg. meas. function. Since  $\mu_3$  can be written in this form,  $\mu_3 \ll \nu$  and

$$\frac{d\mu_3}{d\nu} = \frac{d\mu_1}{d\nu} + \frac{d\mu_2}{d\nu}$$

where this is unique a.e. ( $\nu$ ) by the uniqueness of the Radon-Nikodym derivative.

(d)  $\mu \ll \nu$  and  $\mu \perp \nu$ . Show  $\mu$  is zero measure.  
 $\mu \perp \nu \Rightarrow \exists$  some  $B \in \mathcal{F}$  s.t.

$$\mu(B) = 0 \quad \text{and} \quad \nu(B^c) = 0,$$

but  $\mu \ll \nu \Rightarrow$

$$\nu(B^c) = 0 \Rightarrow \mu(B^c) = 0.$$

Hence,

$$\mu(B) = \mu(B^c) = 0$$

$$\Rightarrow \mu(B \cup B^c) = \mu(B) + \mu(B^c) = 0 = \mu(\Omega).$$

So, for any  $A \in \mathcal{F}$ ,

$$\mu(\emptyset) \leq \mu(A) \leq \mu(\Omega)$$

$$\Rightarrow 0 \leq \mu(A) \leq 0$$

$$\Rightarrow \mu(A) = 0.$$

Thus,  $\mu$  is the zero measure.

(e)  $\mu_1 = \text{lognormal}(5, 6)$ ,  $\mu_2 = \text{Geometric}(0.3)$ ,  $\nu = \text{Poisson}(8)$   
 on  $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ . Want to find the Lebesgue decomposition of  $\mu_3$  wrt  $\nu$  and R-N derivative

$\frac{d\mu_3}{d\nu}$ . Consider that

$$\mu_3(A) = \mu_1(A) + \mu_2(A) = \int_A f_1 d\mu_1 + \int_A f_2 d\mu_2$$

$\swarrow$  Lebesgue meas       $\swarrow$  counting measure  
 $\nu(A) = \int_A f_3 d\mu_3$



where

$$f_1 = \frac{1}{\sqrt{2\pi}(6)} \frac{\exp\{-(\log(x) - 5)^2\} / 2(6)^2}{x} \mathbb{I}_{[0, \infty)}(x)$$

$$f_2 = 0.3(0.7)^i \mathbb{I}_A(i) \quad i \in [0, 1, \dots)$$

$$f_v = \frac{e^{-(8)} 8^i}{i!} \mathbb{I}_A(i) \quad i \in [0, 1, \dots)$$

Let us decompose  $\mu_3$  as

$$\mu_a = \mu_2$$

$$\mu_s = \mu_1.$$

$$\begin{aligned} \text{where } \mu_3 &= \mu_1 + \mu_2 \\ &= \mu_s + \mu_a. \end{aligned}$$

Show  $\mu_a \ll \nu$  ( $\mu_2 \ll \nu$ )

Since we can write  $\mu_a = \mu_2$  as an integral

$$\mu_a(A) = \int f_2 d\mu_c \quad \forall A \in \mathcal{F}$$

where  $f_2$  is a nonnegative measurable function, it follows that  $\mu_a \ll \mu_c$ . Similarly,

$$\nu(A) = \int f_v d\mu_c \Rightarrow \nu \ll \mu_c.$$

Additionally, since  $\nu \ll \mu_c$  and  $\frac{d\nu}{d\mu_c} = f_v > 0$  a.e. ( $\mu_c$ ), it follows that  $\mu_c \ll \nu$ . Thus,

$$\mu_a \ll \mu_c, \mu_c \ll \nu \Rightarrow \mu_a \ll \nu. \quad \checkmark$$

Now, let  $B = \mathbb{N}$ . Then,

$$\mu_1(B) = \mu_1(\mathbb{N}) = 0 \quad \text{and}$$

$$\nu(B^c) = \nu(\mathbb{N}^c) = 0.$$

Hence,  $\mu_1 \perp \nu \Rightarrow \mu_5 \perp \nu$ . ✓

Thus,

$\mu_3$  decomposed as  $\mu_a = \mu_2$  and  $\mu_s = \mu_1$  is the Lebesgue decomposition of  $\mu_3$  wrt  $\nu$ .

Now, to find the R-N derivative

$$\frac{d\mu_{3a}}{d\nu} = \frac{d\mu_2}{d\nu}.$$

Consider that since  $\nu \ll \mu_c$  and  $\frac{d\nu}{d\mu_c} > 0$  a.e. ( $\mu_c$ ), a proposition says that

$$\frac{d\mu_c}{d\nu} = \left( \frac{d\nu}{d\mu_c} \right)^{-1}.$$

Furthermore,  $\mu_2 \ll \mu_c$  and  $\mu_c \ll \nu$ , so

$$\begin{aligned} \frac{d\mu_2}{d\nu} &= \frac{d\mu_2}{d\mu_c} \cdot \frac{d\mu_c}{d\nu} = \frac{d\mu_2}{d\mu_c} \cdot \left( \frac{d\nu}{d\mu_c} \right)^{-1} \\ &= f_1 \cdot f_\nu^{-1} = \frac{f_1}{f_\nu}. \quad \checkmark \end{aligned}$$